

Optimisation report — ludos_panels_n4_optim

Run

Generated 2026-09-10 16:43 — output folder: exports/optim/ludos_panels_n4_optim
Method: differential_evolution budget: 500 evaluations performed: 511 elapsed: 538 s backend: GPU
Decision variables: 6 wall distances: 50 cm eval grid: 100² over 100 cm, 3000 rays/pixel

Key results

	initial	best #1	best #2
Score (lower is better)	1.3411	1.2101	1.2114
Uniformity U0 (u0)	65.9 %	79.0 %	78.9 %
FOV coverage	100 %	100 %	100 %
E_avg in FOV (mean over walls)	890 lx	383 lx	359 lx
Active LEDs	8	8	8
Drivers	2	2	2
Total current	1.2 A	1.2 A	1.2 A
E_min / E_max (last wall)	586 / 1,335 lx	302 / 642 lx	283 / 606 lx
ΔEV (stops)	1.19	1.09	1.10
normal: E_avg	890 lx	383 lx	359 lx
normal: VIO ≥120 lx	0 %	0 %	0 %
penalty: coverage	0.0000	0.0000	0.0000
penalty: normal.vio	1.0000	1.0000	1.0000

Best design improves the score by 9.8 % over the initial configuration (1.3411 → 1.2101).

The 'initial' column is the optimiser's starting point x_0 : the loaded scene when an existing design is refined, or the centre of the search box for a generated layout. The two best designs differ in at least one variable by more than 2 % of its range.

Problem definition

Scene / evaluation settings

Main camera: position (10, -4) cm, pitch 0°, FOV 75° × 60°
Emission: default 168 lm/LED, focus factor 0
Driver model: 6 V, 180 lm/W → 1,080 lm/A, max 13 A per LED, 4 LED(s) per driver
VIO cameras: at [10, 0, 0] cm, cam1 pitch 45° yaw 0°, cam2 pitch -45° yaw 0°, long-side FOV 170° (landscape); VIO wall at 300 cm, 1200 cm wide, 40² cells

Operating modes

- normal: flux × 1; ≥ 50 % of VIO FOV above 120 lx (w=1)

Objective

metric = u0, E_min percentile = 2 %, coverage weight = 1

Constraints (active)

none

Decision variables

variable	min	max	int	initial	best #1	best #2
group0.dx	-10	10		0	-6.621	-8.121
group0.dy	-10	10		0	-4.615	-3.882
group0.dz	-10	10		0	-1.682	1.501
group0.drot_x	-30	30		0	1.97	-0.6854
group0.drot_y	-30	30		0	-1.097	2.211
group0.drot_z	-30	30		0	23.01	22.97

Method & formulas

How a design is scored

Each candidate decision vector x is decoded into a regular saved configuration (same JSON schema as configs/*.json), the scene is built with the exact same code as the UI, and the wall illuminance E (lux) is ray-traced on a grid at every wall distance d in the list above. Only the cells inside the main-camera footprint (the FOV trapezoid, drawn in white on the images) are kept; cells with $E = 0$ are ignored for the uniformity metrics but counted for coverage.

Uniformity metrics over the lit FOV cells:

$$E_{min} = P_2(E), \quad E_{avg} = \bar{E}, \quad E_{max} = \max(E)$$

$$U_0 = \frac{E_{min}}{E_{avg}}, \quad U_1 = \frac{E_{min}}{E_{max}}, \quad CV = \frac{\sigma_E}{E_{avg}}$$

E_{min} uses the 2-th percentile of lit cells so a single Monte-Carlo-starved cell cannot dominate the metric.

Score (minimised):

$$S(x) = \frac{1}{N_d} \sum_d [1 - U_0]_d + 1 \left(1 - \min_d \text{cov}_d \right) + \sum_k P_k(x)$$

cov_d is the fraction of FOV cells receiving any light at distance d . Every constraint is a soft penalty $P_k \geq 0$ added to the score (no hard infeasibility), which keeps the landscape usable for population-based global search.

Active penalty terms:

$$\text{mode normal: } E^{(normal)} = sE, \quad s = 1$$

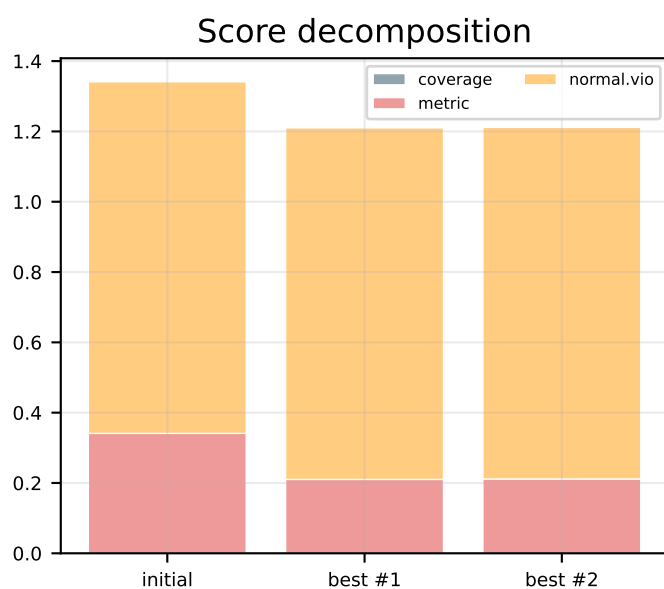
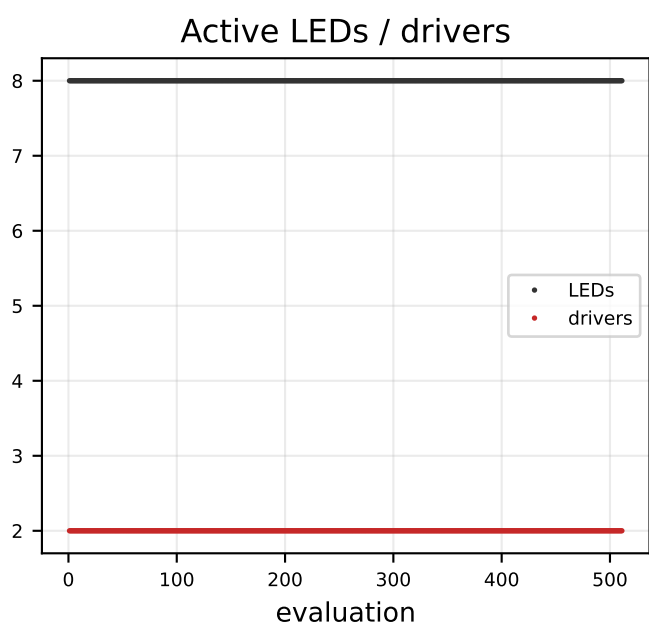
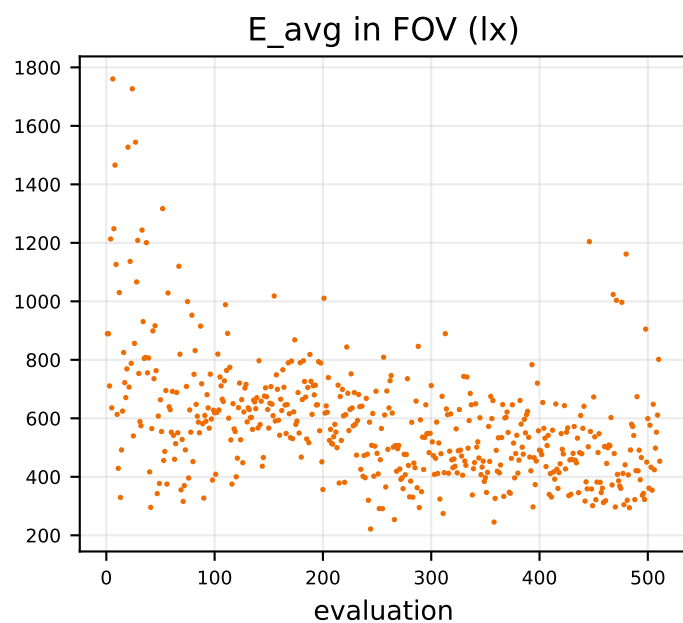
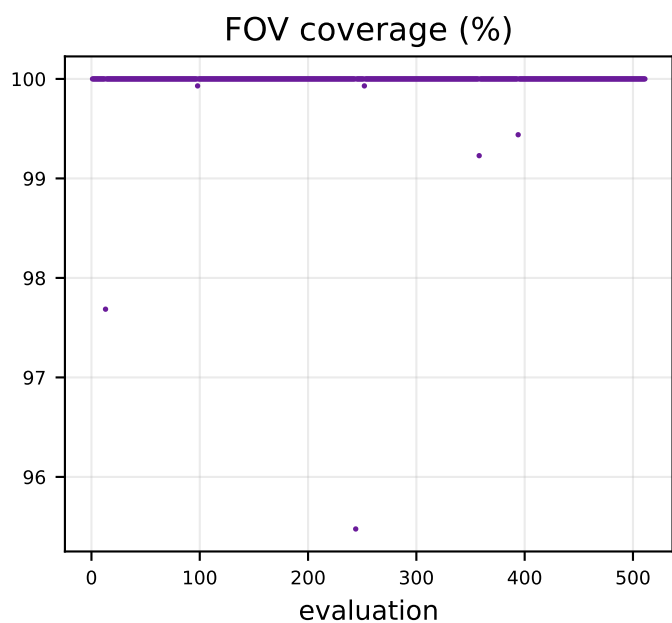
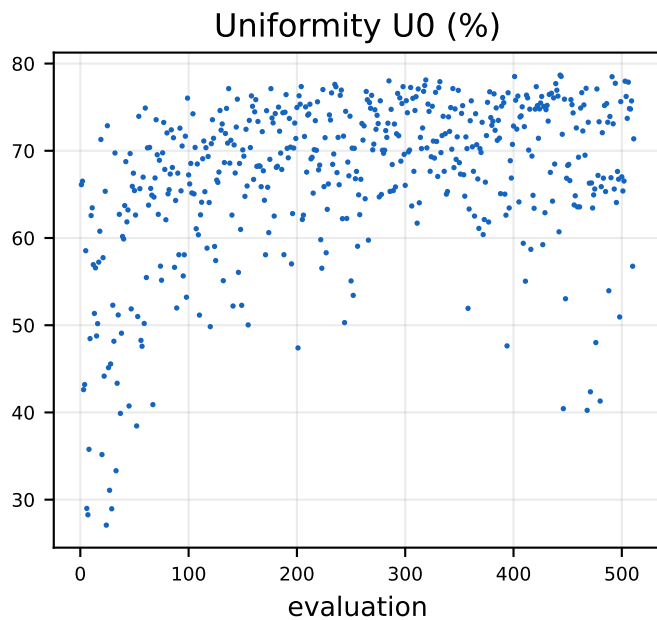
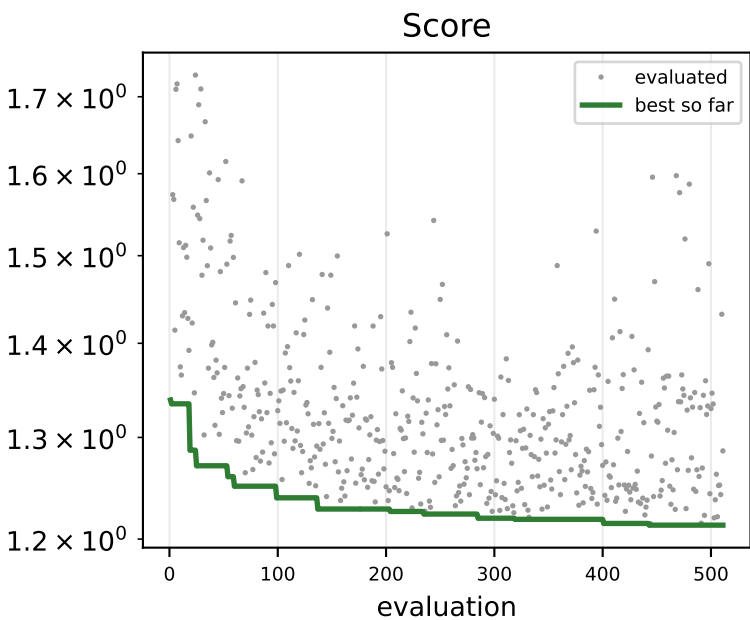
$$P_{normal, vio} = 1 \max\left(0, 1 - \frac{f_{vio}}{0.5}\right), \quad f_{vio} = \frac{\#\{E^{(normal)} \geq 120 \text{ on VIO FOV}\}}{\#\text{VIO FOV cells}}$$

Modes share the geometry and differ only by drive current; since luminous flux is linear in current ($\Phi = I \cdot V \cdot \eta$), the traced grid is rescaled per mode instead of re-traced. $\bar{I}$ is the mean design current implied by the LED flux in the decoded configuration.

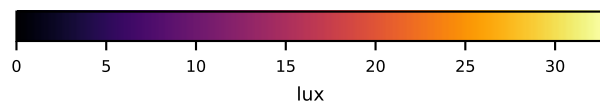
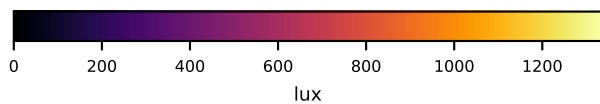
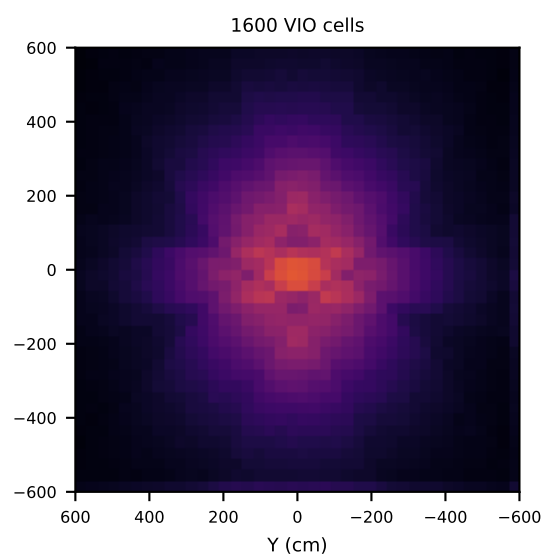
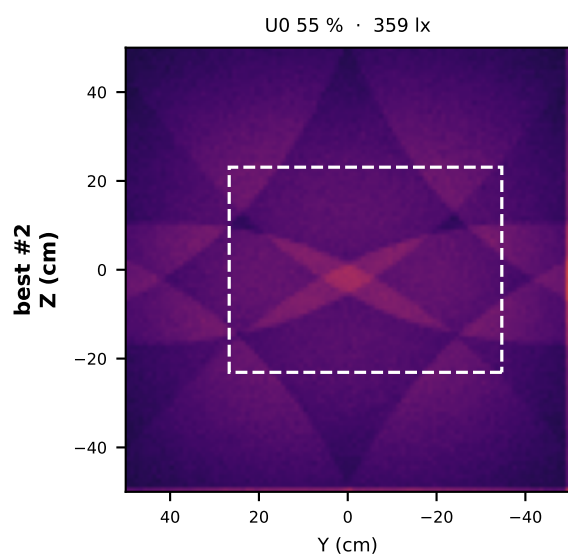
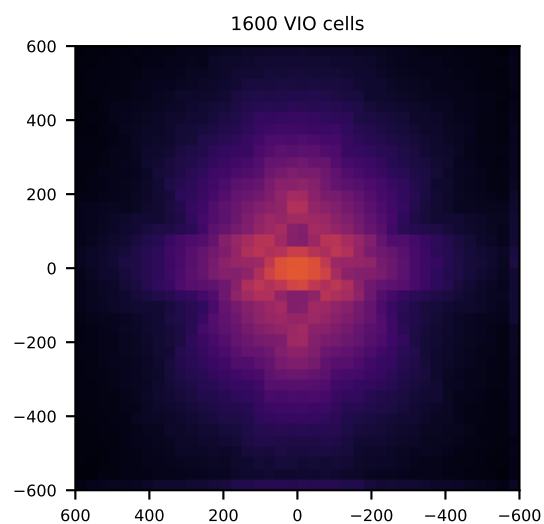
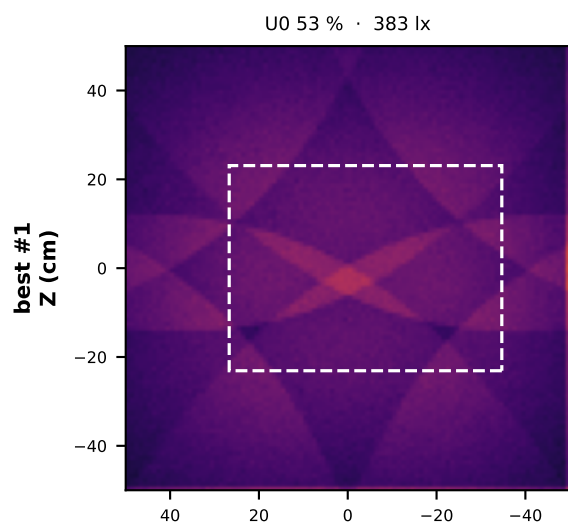
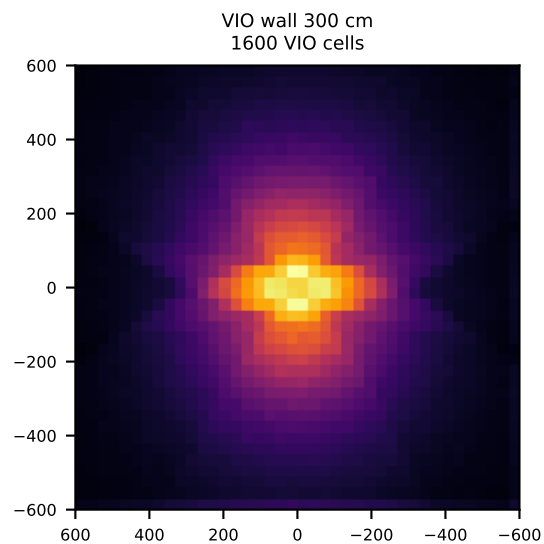
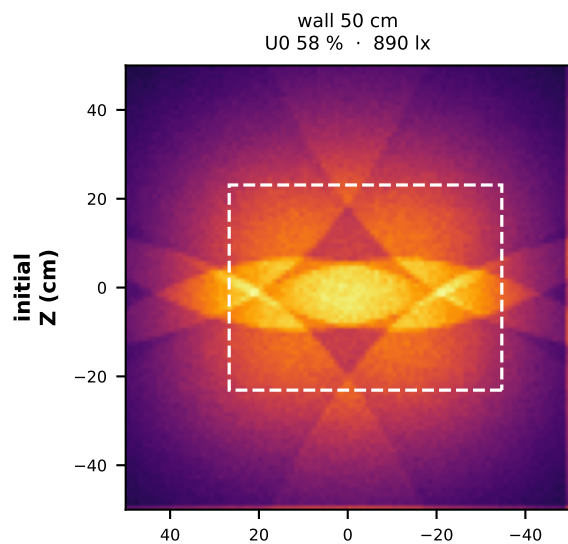
Optimiser:

Differential evolution (scipy) samples a population inside the bounds, recombines candidates and keeps improvements; integer variables (on/off states, row/column counts) are rounded before decoding. Nelder-Mead and random search are local/simple alternatives used for refinement. The best design is exported after every improvement, so a stopped run is never lost.

Convergence

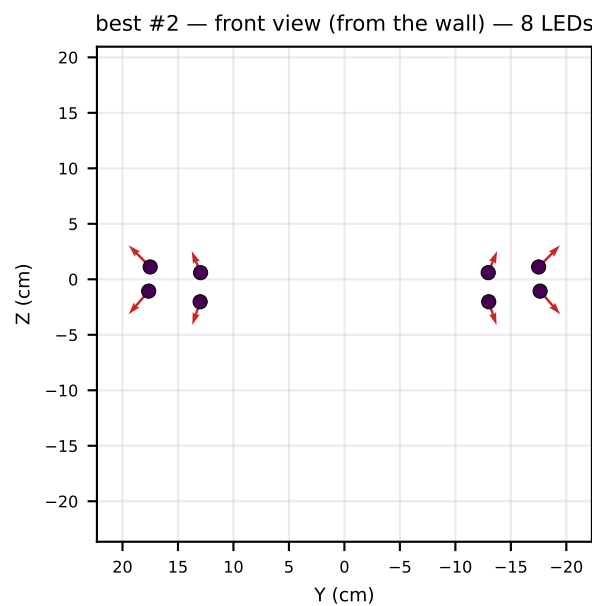
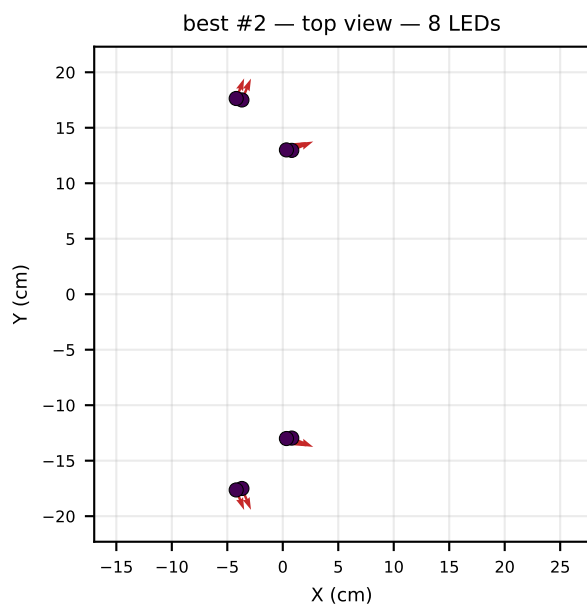
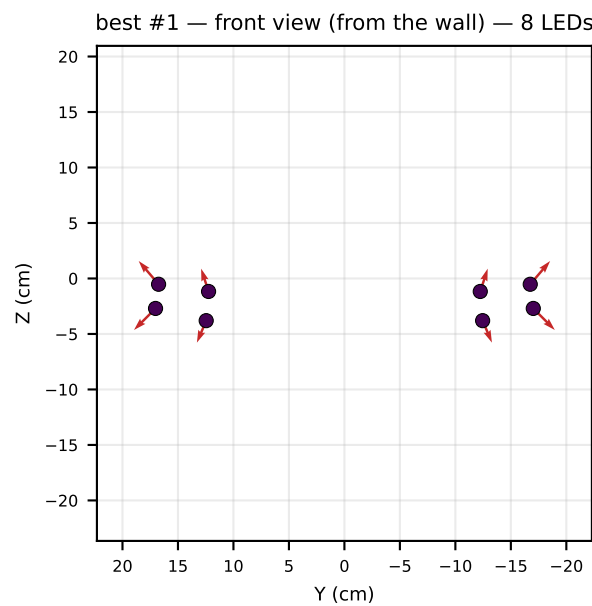
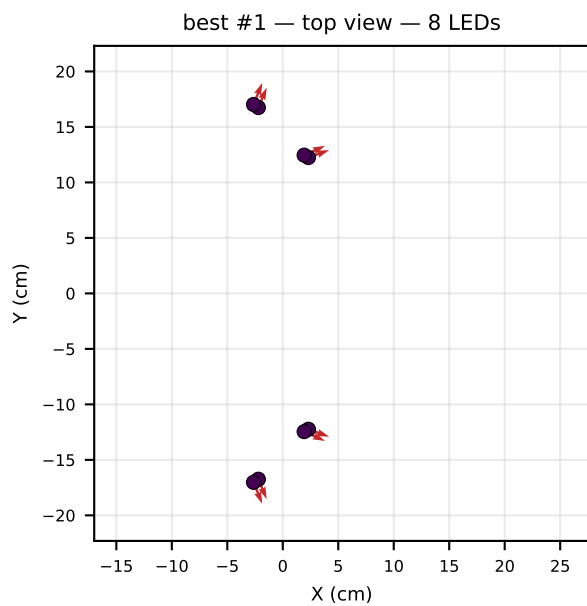
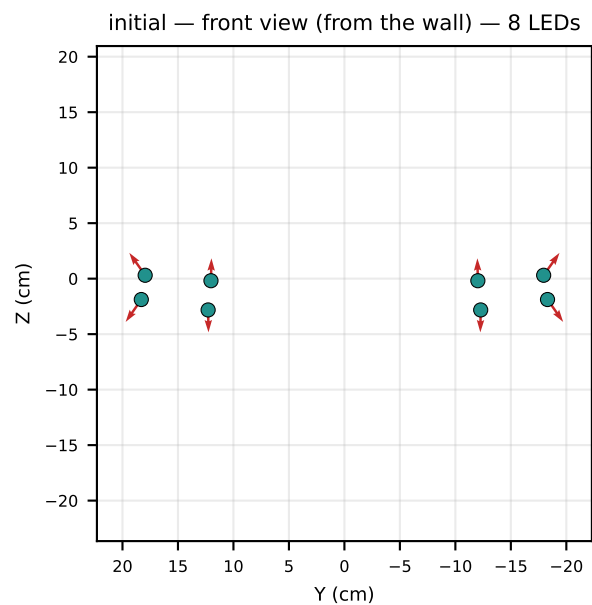
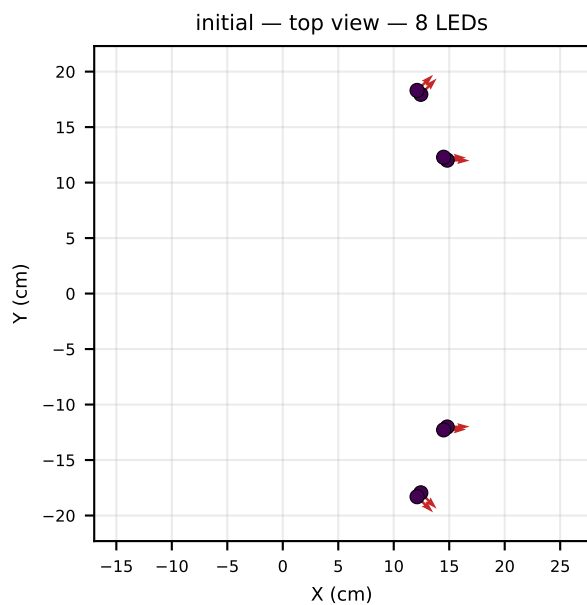


Wall illuminance — initial vs. best designs

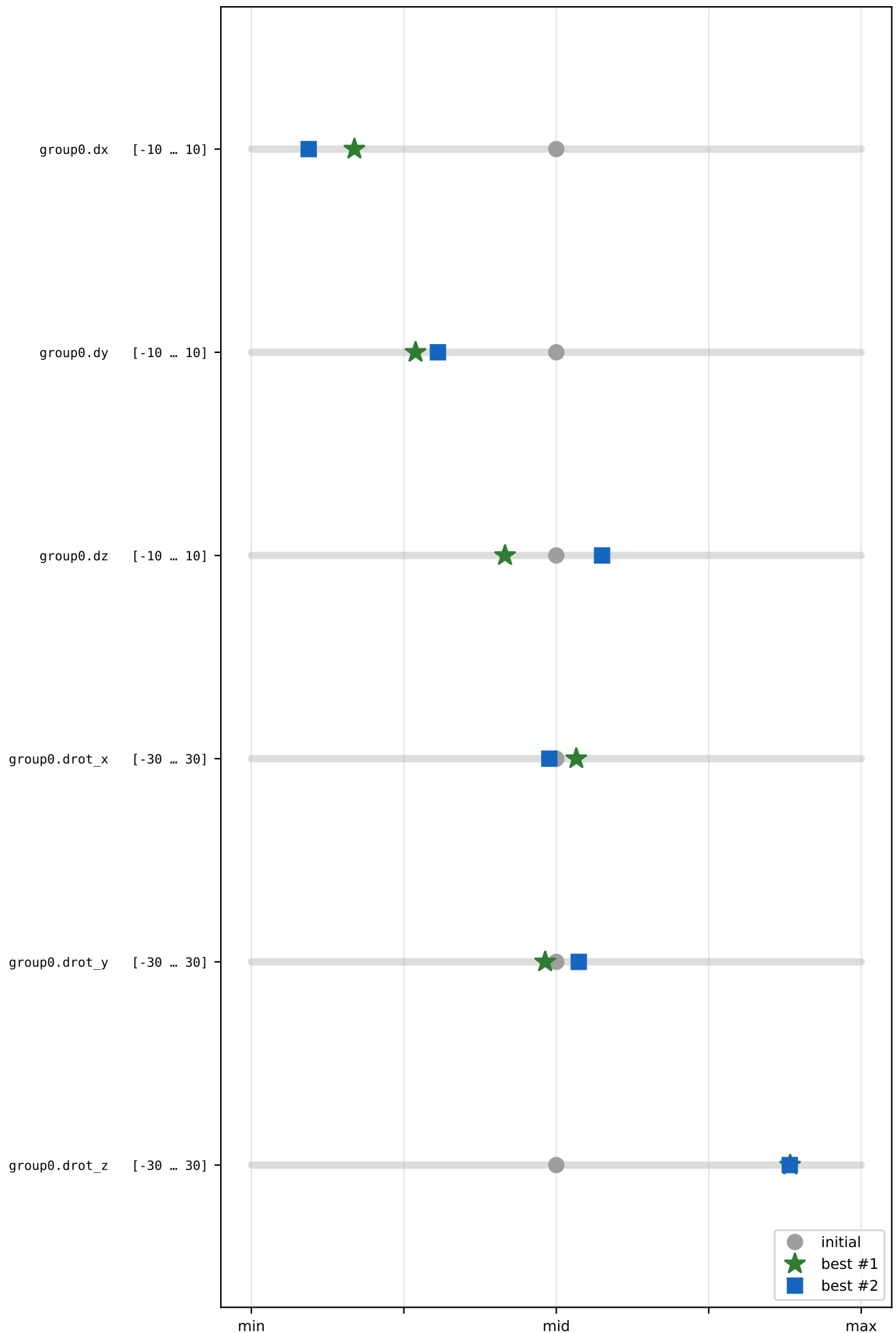


Dashed white: main-camera FOV footprint (metrics are computed inside it). Cyan: VIO camera footprint. Colour scale is shared per column.

LED layouts (active LEDs, arrows = beam axis)

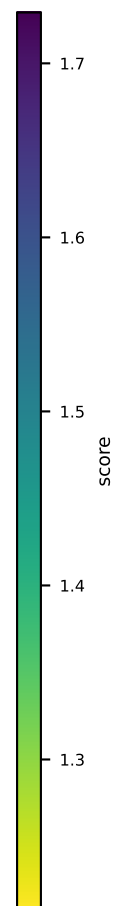
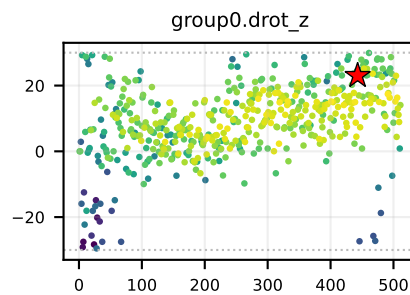
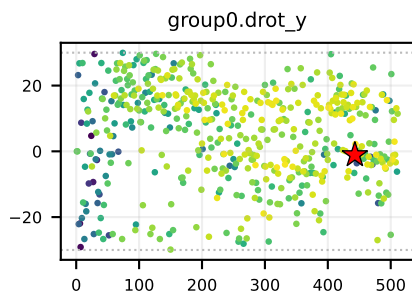
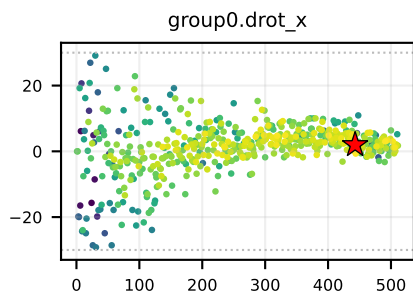
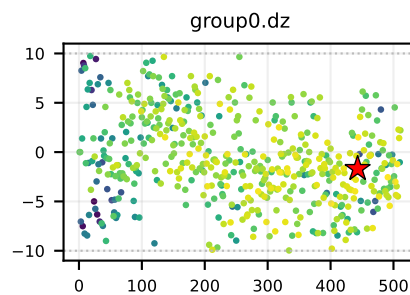
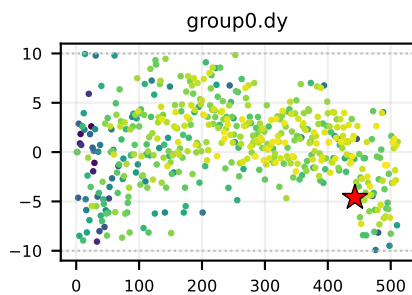
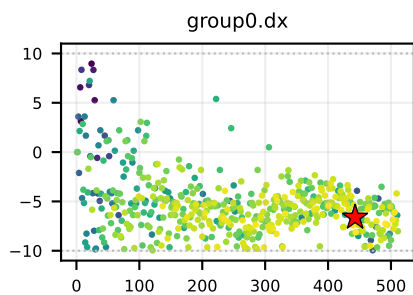


Optimised variables — position inside the search bounds



Values sitting on a bound suggest the search box may be too tight for that variable.

Optimised variables — explored values (colour = score, ★ = best)



Files

best_config.json — best design, loadable in the UI (Project → Load) or via --evaluate

best2_config.json / initial_config.json — the other two designs compared in this report

history.csv — every logged evaluation with score, metrics and all variable values

summary.json — run metadata, best evaluation and best decision vector

initial: score=1.3411 U=65.9% cov=100% Eavg=890lx LEDs=8 drv=2 I=1.2A normal[Eavg=890lx VIO=0%]
[normal.vio=1.000]

best #1: score=1.2101 U=79.0% cov=100% Eavg=383lx LEDs=8 drv=2 I=1.2A normal[Eavg=383lx VIO=0%]
[normal.vio=1.000]

best #2: score=1.2114 U=78.9% cov=100% Eavg=359lx LEDs=8 drv=2 I=1.2A normal[Eavg=359lx VIO=0%]
[normal.vio=1.000]